CMPS 350 – Web Development Fundamentals

Practical Midterm Exam

- The exam duration is 120 minutes. So, read the exam questions carefully and plan your time accordingly.
- Push your code to GitHub regularly (at least every 30 minutes) to avoid unpleasant surprises, as your computer might hang!
- The Exam is an open book. In case of plagiarism, both parties will receive 0 points. Hence do not share or receive any code from anyone.
- Once you complete the exam, you should:
    - Add a screenshot for each question to the provided testing sheet. ○ Push your code and testing sheet to your GitHub repo under **`midterm`** subfolder. ○ Demo your work before leaving the Exam.

Gaming Store App

In this exam, you will apply your frontend web development skills to build a gaming web app that displays gaming products in a grid. The UI design is provided in the figure below and a demo of the app will be shown at the start of the exam.

Implementation Tasks:

1. **[30 pts]** When the page loads **Get gaming products** from the following API and display them in a grid of cards as shown in the image below. The game product card should display the `Product Image`, `Product Name`, `Price`, and `Availability`. **The cards should be responsive.**

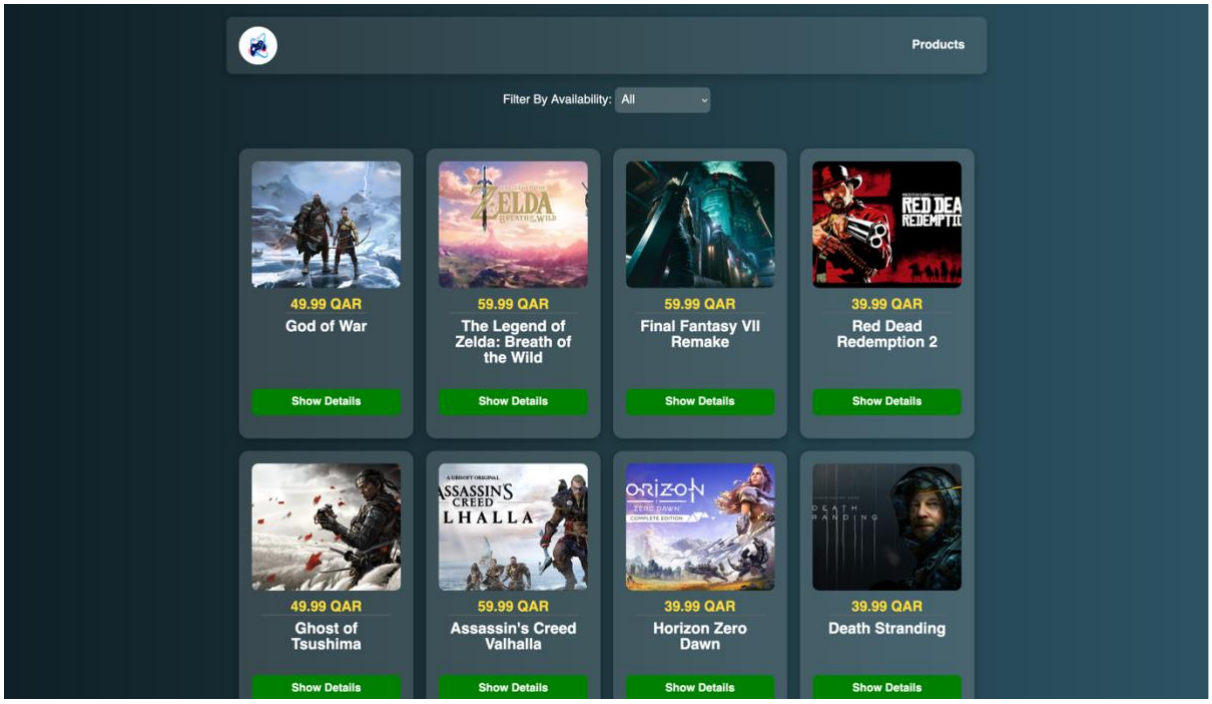

Figure 1 Products page

2. **[20 pts] Caching Products in the Local Storage**
   Improve your app performance by caching the products data in the local storage. When you get the products data, save them to the local storage. Subsequent visits

should first read the data from the local storage. Then only fetch from the remote Web API if the products data is not available in the local storage.

3. **[20 pts] Filter Products by Availability :** Implement a dropdown filter to allow users to sort products based on:

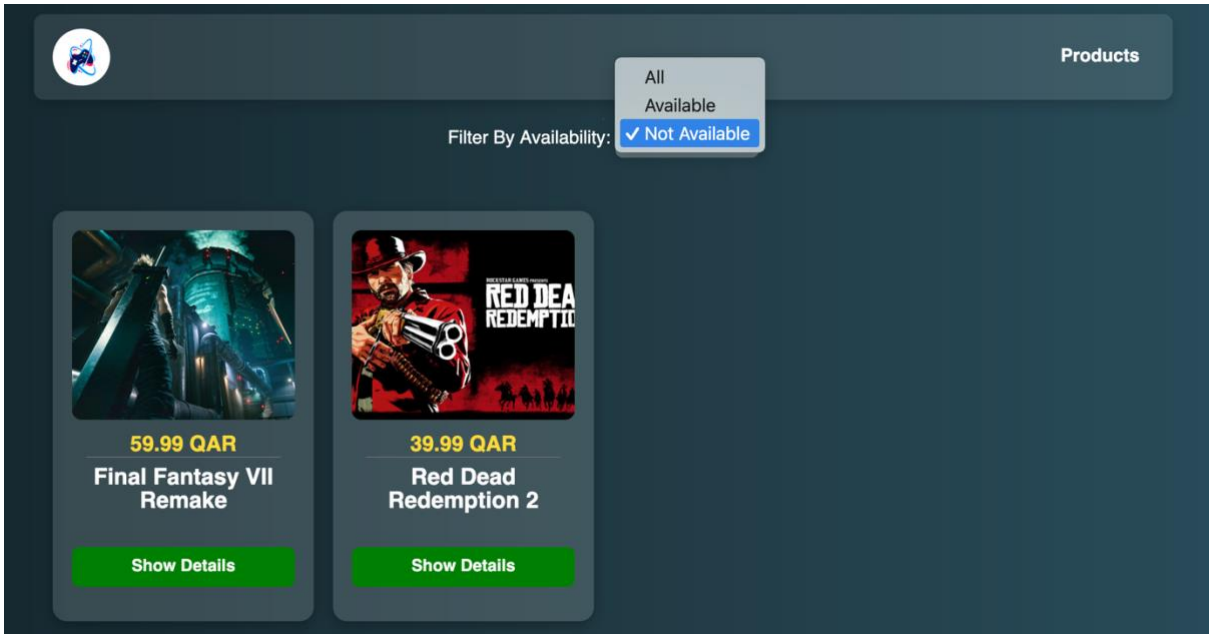

Figure 2 Products page after filtering

4. **[30 pts] Show Details View**
    - Add a "Show Details" button on each product card.
    - Clicking "Show Details" should:
    - Replace the grid view with a single expanded card displaying all product details (including description).
        - Show a "Back" button that returns the user to the grid view.
        - Show a "Delete" button to remove the product from the details view and local storage.

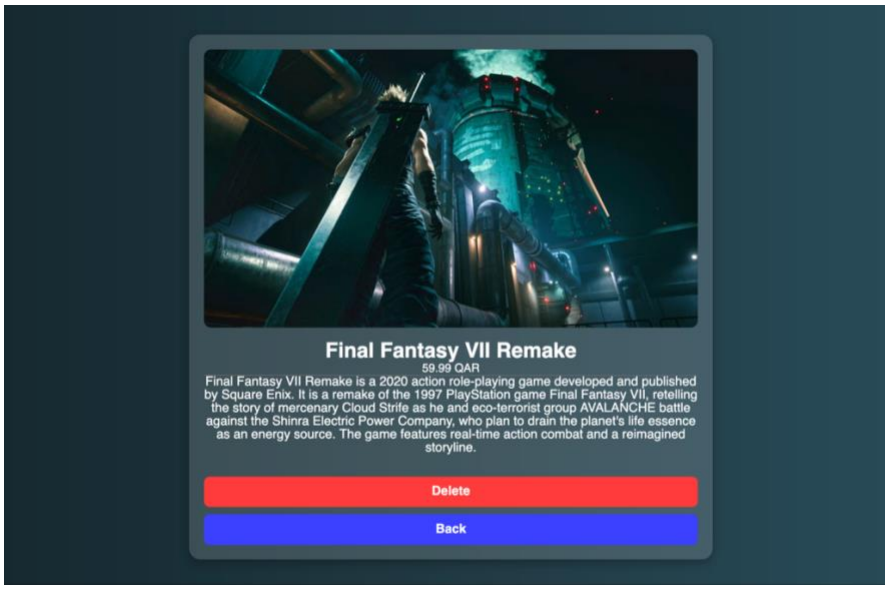

Figure 3 Details page